The hierarchy of n-types is cumulative: an n-type is also an (n+1)-type

The Univalent Foundations Program

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Editorial note. The counted statement is the author's ‘thm:hlevel-cumulative‘ (source lines 113-116, Theorem 7.1.7); the proof environment begins after one blank line, so the pairing is by adjacency.

Editorial note. The retained context is the author's own remark that the preceding theorem fails at n = -2, which is the boundary at which the cumulativity theorem has to start.

Editorial note. The retained counter values (chapter 7, section 1, theorem offset 6) make the retained theorem print as Theorem 7.1.7, matching the original book numbering.

Editorial note. No retained line refers to a label defined outside the retained span, so the package declares no external reference and no bibliography.

Editorial scope. One original result of the HoTT Book, counted once: cumulativity of the hierarchy of n-types (Theorem 7.1.7).

Allowed standard prerequisites (editorial).

- n-types and contractibility. X is a (-2)-type if it is contractible and an (n+1)-type if all its path spaces are n-types.. The base case at source lines 121-128 is stated for contractible types and the induction step at lines 130-132 is exactly the path-space definition.
- Contractibility as a pair of centre and contraction. A contractible type comes with a centre a_0 and, for every x, a path contr_x : a_0 = x.. The centre and the contraction are what build the candidate centre contr_x ‘ct‘ opp(contr_y) at source line 124.
- Path induction. To prove a family of statements about p : x = y for all p it suffices to prove it at refl x.. It reduces the contraction obligation at source lines 126-128 to refl x = contr_x ‘ct‘ opp(contr_x).

Note that this theorem fails when $n = -2$: the map $\mathbf{0} \to \mathbf{1}$ is an embedding, but $\mathbf{1}$ is a (-2)-type while $\mathbf{0}$ is not.

Theorem 7.1.7. *The hierarchy of n-types is cumulative in the following sense: given a number $n \geq -2$, if X is an n-type, then it is also an $(n+1)$-type.*

Proof. We proceed by induction on n.

For $n = -2$, we need to show that a contractible type, say, A, has contractible path spaces. Let $a_0 : A$ be the center of contraction of A, and let $x, y : A$. We show that $x =_A y$ is contractible. By contractibility of A we have a path $\mathsf{contr}_x \cdot \mathsf{contr}_y{}^{-1} : x = y$, which we choose as the center of contraction for $x = y$. Given any $p : x = y$, we need to show $p = \mathsf{contr}_x \cdot \mathsf{contr}_y{}^{-1}$. By path induction, it suffices to show that $\mathsf{refl}_x = \mathsf{contr}_x \cdot \mathsf{contr}_x{}^{-1}$, which is trivial.

For the inductive step, we need to show that $x =_X y$ is an $(n+1)$-type, provided that X is an $(n+1)$-type. Applying the inductive hypothesis to $x =_X y$ yields the desired result. □